# STA106_Midterm2_Project_RCode

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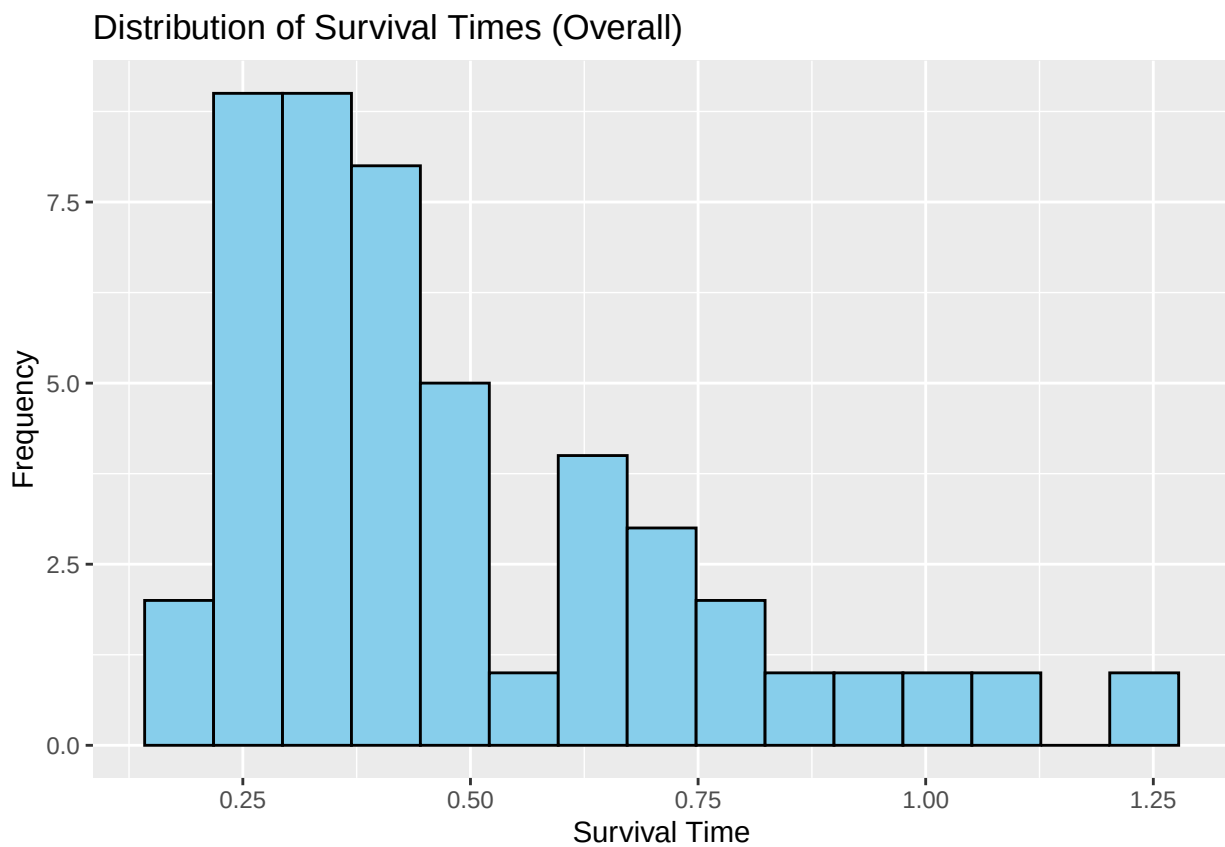
## Load the data

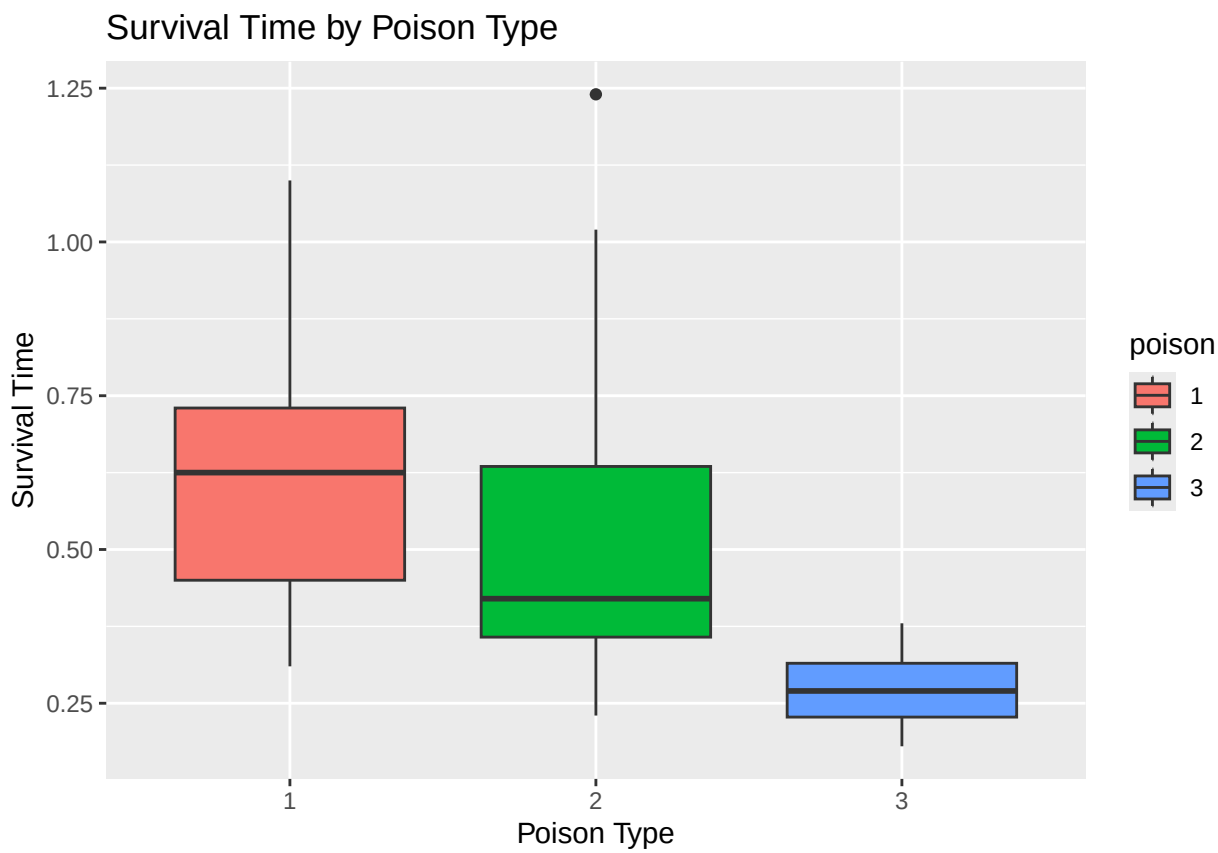
```r
df <- read.csv("poisons.csv")

# poison and treatment are factors
df$poison <- factor(df$poison)
df$treat <- factor(df$treat)
```

## Figures

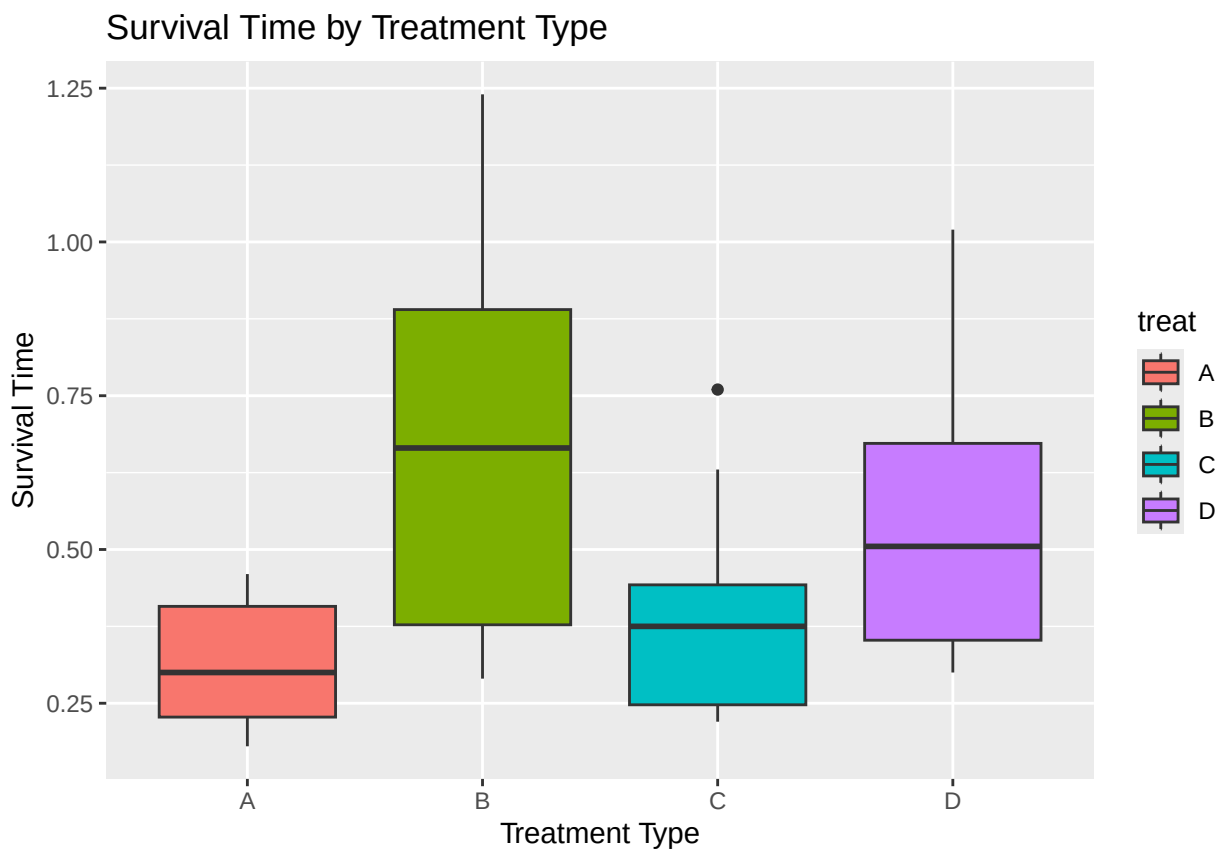
```r
# Figure 1: Overall Histogram of Survival Time
ggplot(df, aes(x = time)) +
  geom_histogram(fill = "skyblue", color = "black", bins = 15) +
  labs(title = "Distribution of Survival Times (Overall)",
       x = "Survival Time",
       y = "Frequency")
```



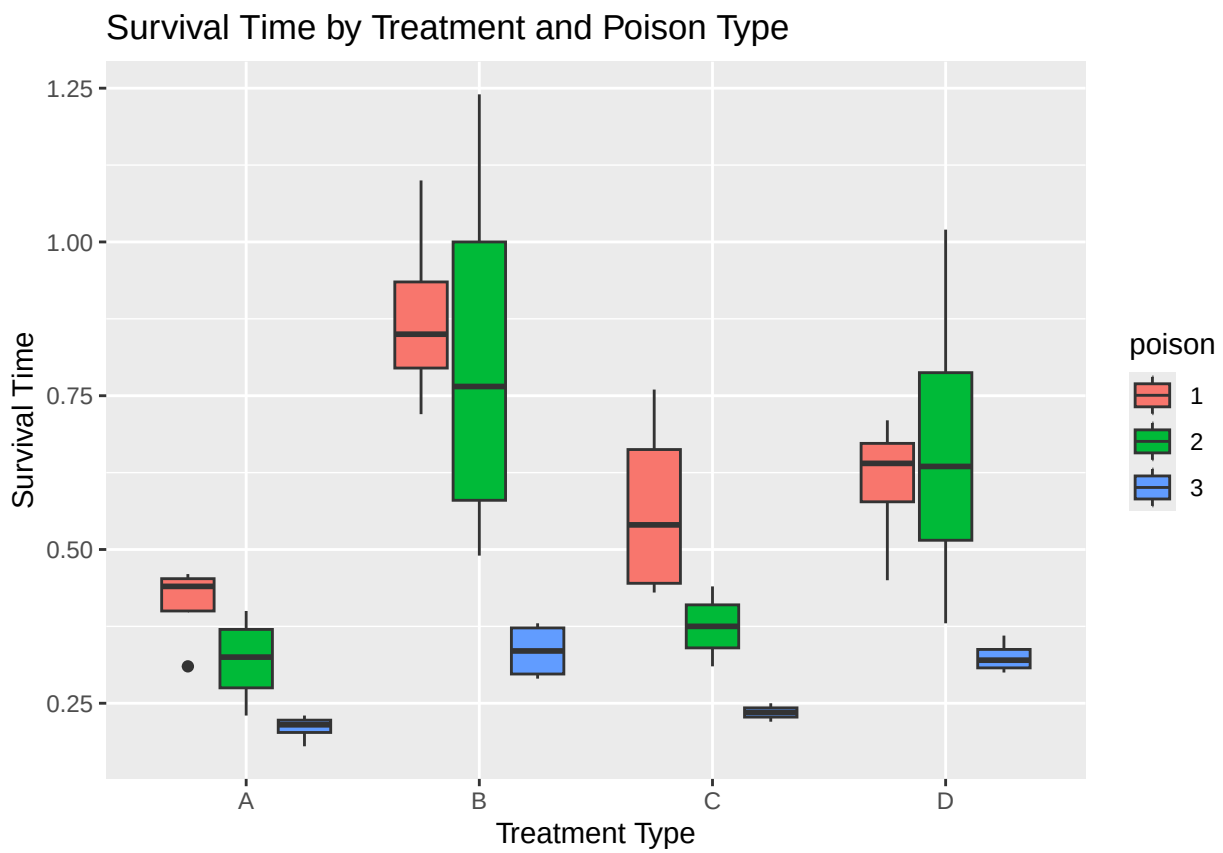

```r
# Figure 2: Boxplot by Poison Type
ggplot(df, aes(x = poison, y = time, fill = poison)) +
  geom_boxplot() +
  labs(title = "Survival Time by Poison Type",
       x = "Poison Type",
       y = "Survival Time")
```




```r
# Figure 3: Boxplot by Treatment Type
ggplot(df, aes(x = treat, y = time, fill = treat)) +
  geom_boxplot() +
  labs(title = "Survival Time by Treatment Type",
       x = "Treatment Type",
       y = "Survival Time")
```




```r
# Figure 4:survival Time by Treatment and Poison Type
ggplot(df, aes(x = treat, y = time, fill = poison)) +
  geom_boxplot() +
  labs(title = "Survival Time by Treatment and Poison Type",
       x = "Treatment Type",
       y = "Survival Time")
```


Survival Time by Treatment and Poison Type
1.25
1.00
0.75
0.50
0.25
Survival Time
A
B
C
D
Treatment Type
poison
1
2
3

## Summaries

```r
# 1. Grouped Summary (Poison & Treatment)
summary_stats <- df %>%
  group_by(poison, treat) %>%
  summarise(
    Mean_Time = mean(time, na.rm = TRUE),
    SD_Time = sd(time, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  )

# 2. Summary by Poison Type
poison_summary <- df %>%
  group_by(poison) %>%
  summarise(
    Mean_Time = mean(time, na.rm = TRUE),
    SD_Time = sd(time, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  )

# 3. Summary by Treatment Type
treat_summary <- df %>%
  group_by(treat) %>%
  summarise(
    Mean_Time = mean(time, na.rm = TRUE),
    SD_Time = sd(time, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  )

# 4. Overall Summary
overall_stats <- df %>%
  summarise(
    Mean_Time = mean(time, na.rm = TRUE),
    SD_Time = sd(time, na.rm = TRUE),
    n = n()
  )
```

## Tables

```r
# Table 1: Overall
knitr::kable(
  overall_stats,
  booktabs = TRUE,
  digits = 4,
  caption = "Overall Summary Statistics",
  col.names = c("Overall Mean Time", "Overall Std. Deviation", "Total Sample Size")
) %>%
  kableExtra::kable_styling(latex_options = "hold_position", full_width = FALSE)
```

Table 1: Overall Summary Statistics

| Overall Mean Time | Overall Std. Deviation | Total Sample Size |
|---|---|---|
| 0.4794 | 0.2529 | 48 |

```r
# Table 2: Grouped
knitr::kable(
  summary_stats,
  booktabs = TRUE,
  digits = 4,
  caption = "Summary Statistics by Poison and Treatment",
  col.names = c("Poison", "Treatment", "Mean Time", "Std. Deviation", "Sample Size")
) %>%
  kableExtra::kable_styling(latex_options = "hold_position", full_width = FALSE)
```

Table 2: Summary Statistics by Poison and Treatment

| Poison | Treatment | Mean Time | Std. Deviation | Sample Size |
|---|---|---|---|---|
| 1 | A | 0.4125 | 0.0695 | 4 |
| 1 | B | 0.8800 | 0.1608 | 4 |
| 1 | C | 0.5675 | 0.1567 | 4 |
| 1 | D | 0.6100 | 0.1128 | 4 |
| 2 | A | 0.3200 | 0.0753 | 4 |
| 2 | B | 0.8150 | 0.3363 | 4 |
| 2 | C | 0.3750 | 0.0569 | 4 |
| 2 | D | 0.6675 | 0.2710 | 4 |
| 3 | A | 0.2100 | 0.0216 | 4 |
| 3 | B | 0.3350 | 0.0465 | 4 |
| 3 | C | 0.2350 | 0.0129 | 4 |
| 3 | D | 0.3250 | 0.0265 | 4 |

```r
# Table 3: By Poison Type
knitr::kable(
  poison_summary,
  booktabs = TRUE,
  digits = 4,
```

```r
  caption = "Summary Statistics by Poison Type",
  col.names = c("Poison Type", "Mean Time", "Std. Deviation", "Sample Size")
) %>%
  kableExtra::kable_styling(latex_options = "hold_position", full_width = FALSE)
```

Table 3: Summary Statistics by Poison Type

| Poison Type | Mean Time | Std. Deviation | Sample Size |
|---|---|---|---|
| 1 | 0.6175 | 0.2094 | 16 |
| 2 | 0.5444 | 0.2894 | 16 |
| 3 | 0.2762 | 0.0623 | 16 |

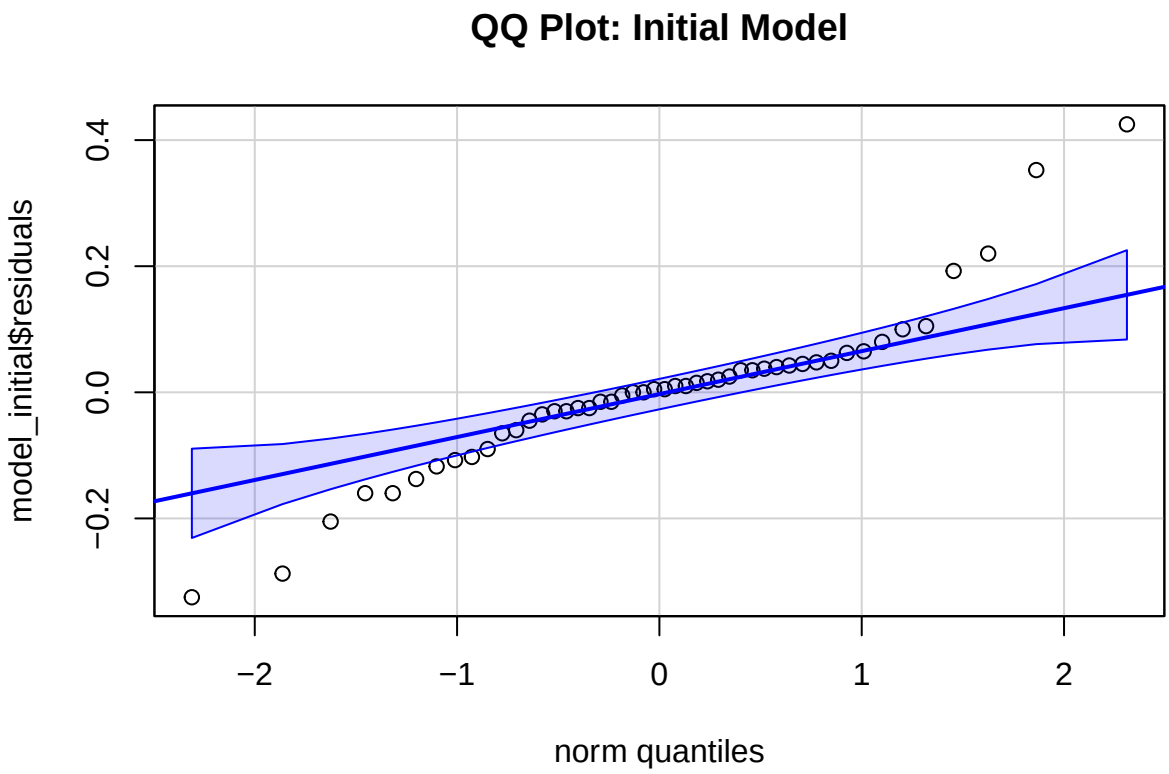
```r
# Table 4: By Treatment Type
knitr::kable(
  treat_summary,
  booktabs = TRUE,
  digits = 4,
  caption = "Summary Statistics by Treatment Type",
  col.names = c("Treatment Type", "Mean Time", "Std. Deviation", "Sample Size")
) %>%
  kableExtra::kable_styling(latex_options = "hold_position", full_width = FALSE)
```

Table 4: Summary Statistics by Treatment Type

| Treatment Type | Mean Time | Std. Deviation | Sample Size |
|---|---|---|---|
| A | 0.3142 | 0.1023 | 12 |
| B | 0.6767 | 0.3208 | 12 |
| C | 0.3925 | 0.1670 | 12 |
| D | 0.5342 | 0.2194 | 12 |

# Diagnosistic

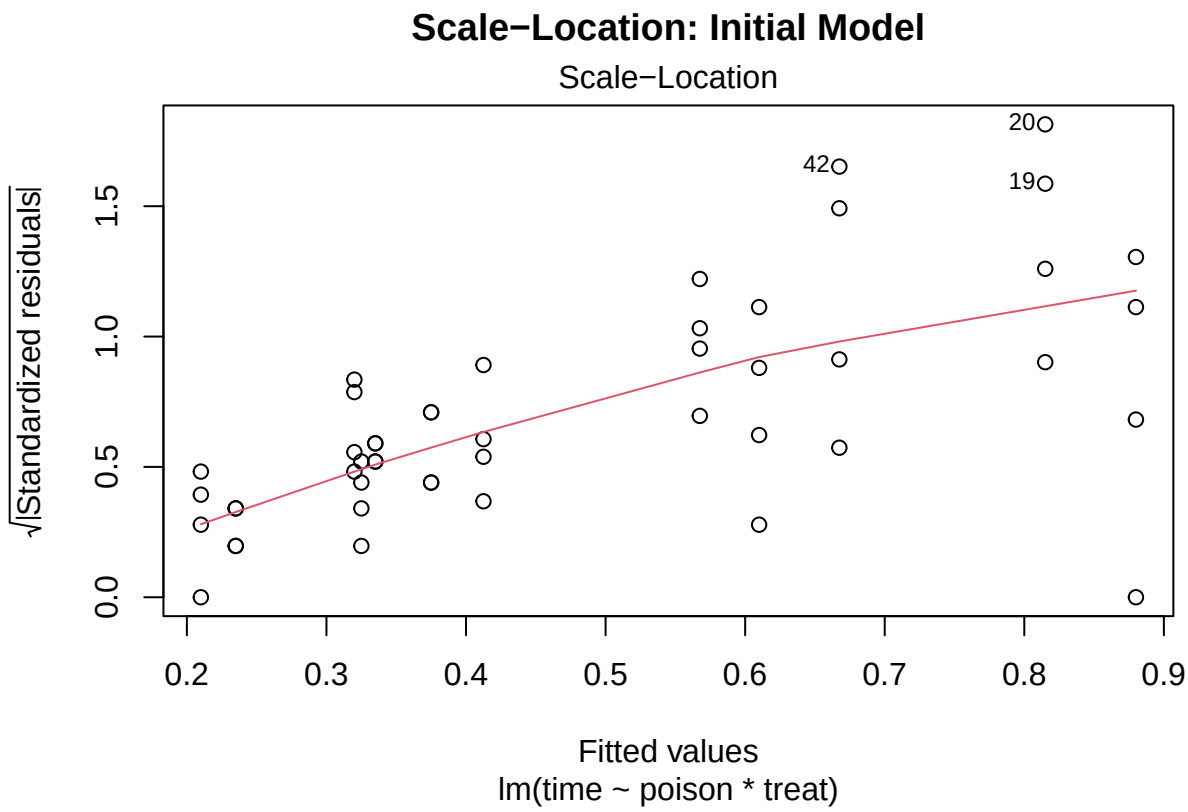
```r
# fit initial interaction model
model_initial <- lm(time ~ poison * treat, data = df)

# 1. Normality Check
car::qqPlot(model_initial$residuals, id = FALSE, main = "QQ Plot: Initial Model")
```




```r
shapiro_initial <- shapiro.test(model_initial$residuals)
print(shapiro_initial)
```

```
##
## Results of Hypothesis Test
## --------------------------
##
## Alternative Hypothesis:
##
## Test Name:                          Shapiro-Wilk normality test
##
## Data:                               model_initial$residuals
##
## Test Statistic:                     W = 0.9122892
##
## P-value:                            0.001606758
```

```r
# 2. Constant Variance Check
plot(model_initial, which = 3, main = "Scale-Location: Initial Model")
```




```r
levene_initial <- leveneTest(model_initial)
print(levene_initial)
```

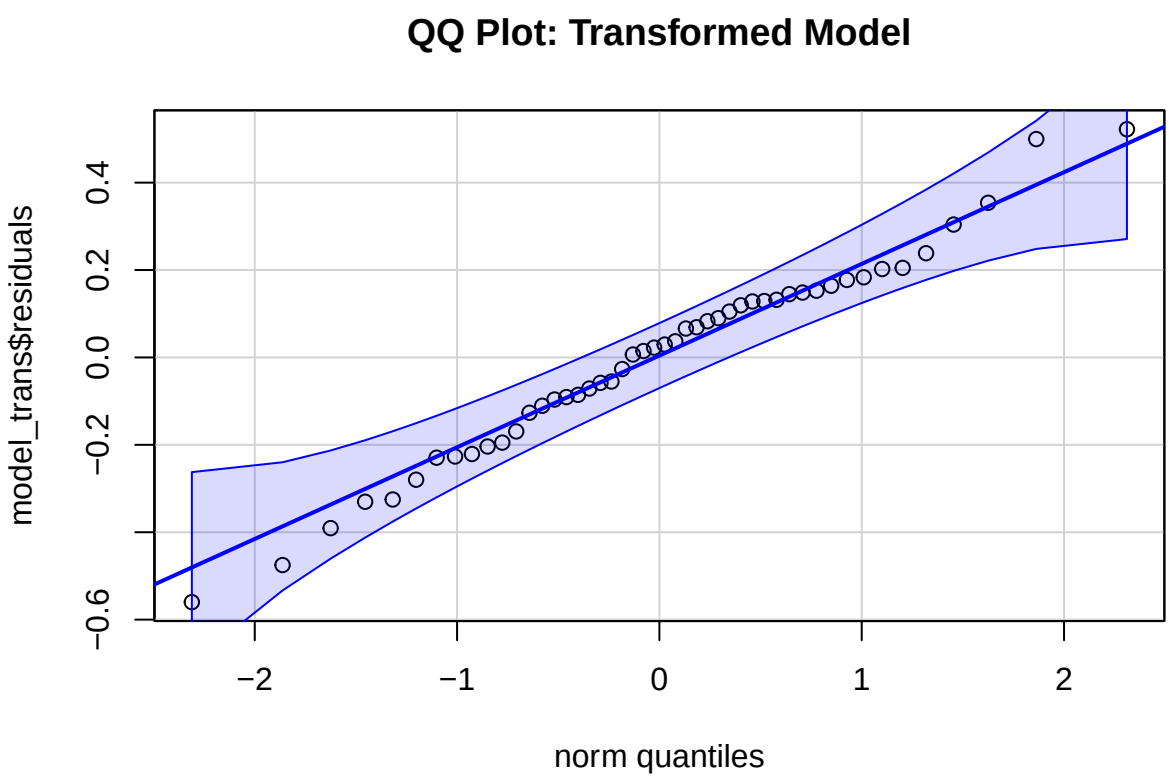
```
##       Df F value       Pr(>F)
## group 11 4.13233 0.0005833475
##       36      NA           NA
```

```r
# 3. Outlier Check with Studentized Residuals and Bonferroni cutoff
nt <- nrow(df)
a <- length(unique(df$poison))
b <- length(unique(df$treat))
alpha <- 0.05
t.cutoff <- qt(1 - alpha / (2 * nt), nt - a * b)
rij <- rstandard(model_initial)
outliers <- which(abs(rij) > t.cutoff)
print(paste("Outliers detected at rows:", paste(outliers, collapse=", ")))
```

```
## [1] "Outliers detected at rows: "
```

```r
# 4. Box-Cox Transformation
# get best lambda using Shapiro-Wilk
L2 <- boxcox(model_initial, objective.name = "Shapiro-Wilk", optimize = TRUE)$lambda

# apply transformation
```

```r
df$time_trans <- (df$time^L2 - 1) / L2

# fit new model with transformed data
model_trans <- lm(time_trans ~ poison * treat, data = df)

# redo diagnostics on transformed model
car::qqPlot(model_trans$residuals, id = FALSE, main = "QQ Plot: Transformed Model")
```




```r
print(shapiro.test(model_trans$residuals))
```

```
##
## Results of Hypothesis Test
## --------------------------
##
## Alternative Hypothesis:
##
## Test Name:                      Shapiro-Wilk normality test
##
## Data:                           model_trans$residuals
##
## Test Statistic:                 W = 0.9839998
##
## P-value:                        0.7496128
```

```r
print(leveneTest(model_trans))
```

```
##       Df  F value     Pr(>F)
## group 11 2.155772 0.04091845
##       36       NA         NA
```

```r
# select final model based on diagnostics
final_model <- model_trans
```

## Analysis

```r
# 1. ANOVA Table (Testing for Interactions and Main Effects)
anova_results <- anova(final_model)
knitr::kable(
  anova_results,
  booktabs = TRUE,
  digits = 4,
  caption = "Two-Way ANOVA Table for Final Model"
) %>%
  kableExtra::kable_styling(latex_options = "hold_position", full_width = FALSE)
```

Table 5: Two-Way ANOVA Table for Final Model

| | Df | Sum Sq | Mean Sq | F value | Pr(>F) |
|---|---|---|---|---|---|
| poison | 2 | 7.3634 | 3.6817 | 54.0489 | 0.0000 |
| treat | 3 | 4.8128 | 1.6043 | 23.5513 | 0.0000 |
| poison:treat | 6 | 0.4389 | 0.0732 | 1.0739 | 0.3962 |
| Residuals | 36 | 2.4522 | 0.0681 | NA | NA |

```r
# 2. Confidence Intervals (4 Pairwise, 2 Contrasts)
# use estimated marginal means for the interaction
emm <- emmeans(final_model, ~ poison * treat)

# --- 4 Pairwise Comparisons ---
# compare treatments within each Poison level (w/ adjusted with Tukey)
pairwise_results <- pairs(emmeans(final_model, ~ treat | poison), adjust = "tukey")
pairwise_df <- as.data.frame(pairwise_results)
knitr::kable(
  pairwise_df,
  booktabs = TRUE,
  digits = 4,
  caption = "Pairwise Comparisons of Treatments within Poison Types (Tukey Adjusted)"
) %>%
  kableExtra::kable_styling(latex_options = c("hold_position", "scale_down"))
```

Table 6: Pairwise Comparisons of Treatments within Poison Types (Tukey Adjusted)

| contrast | poison | estimate | SE | df | t.ratio | p.value |
|---|---|---|---|---|---|---|
| A - B | 1 | -0.8376 | 0.1846 | 36 | -4.5386 | 0.0003 |
| A - C | 1 | -0.3460 | 0.1846 | 36 | -1.8749 | 0.2566 |
| A - D | 1 | -0.4444 | 0.1846 | 36 | -2.4079 | 0.0936 |
| B - C | 1 | 0.4916 | 0.1846 | 36 | 2.6637 | 0.0535 |
| B - D | 1 | 0.3932 | 0.1846 | 36 | 2.1307 | 0.1628 |
| C - D | 1 | -0.0984 | 0.1846 | 36 | -0.5330 | 0.9505 |
| A - B | 2 | -1.0161 | 0.1846 | 36 | -5.5061 | 0.0000 |
| A - C | 2 | -0.2142 | 0.1846 | 36 | -1.1605 | 0.6551 |
| A - D | 2 | -0.8027 | 0.1846 | 36 | -4.3493 | 0.0006 |
| B - C | 2 | 0.8020 | 0.1846 | 36 | 4.3455 | 0.0006 |
| B - D | 2 | 0.2135 | 0.1846 | 36 | 1.1568 | 0.6574 |
| C - D | 2 | -0.5885 | 0.1846 | 36 | -3.1888 | 0.0150 |
| A - B | 3 | -0.5974 | 0.1846 | 36 | -3.2368 | 0.0132 |
| A - C | 3 | -0.1547 | 0.1846 | 36 | -0.8381 | 0.8360 |
| A - D | 3 | -0.5670 | 0.1846 | 36 | -3.0726 | 0.0201 |
| B - C | 3 | 0.4427 | 0.1846 | 36 | 2.3987 | 0.0955 |
| B - D | 3 | 0.0303 | 0.1846 | 36 | 0.1642 | 0.9984 |
| C - D | 3 | -0.4124 | 0.1846 | 36 | -2.2345 | 0.1333 |

```r
# --- 2 Contrasts ---
# contrast 1: Poison 1 vs average of Poison 2 and 3 (across all treatments)
# contrast 2: Treatment A vs Treatment B (across all poisons)
emm_poison <- emmeans(final_model, ~ poison)
emm_treat <- emmeans(final_model, ~ treat)

contrast_list_poison <- list("Poison 1 vs (2+3)/2" = c(1, -0.5, -0.5))
contrast_list_treat <- list("Treat A vs Treat B" = c(1, -1, 0, 0))

contrast_poison_df <- as.data.frame(contrast(emm_poison, contrast_list_poison, adjust = "bonferroni"))
contrast_treat_df <- as.data.frame(contrast(emm_treat, contrast_list_treat, adjust = "bonferroni"))
combined_contrasts <- rbind(contrast_poison_df, contrast_treat_df)

knitr::kable(
  combined_contrasts,
  booktabs = TRUE,
  digits = 4,
  caption = "Custom Contrasts (Bonferroni Adjusted)"
) %>%
  kableExtra::kable_styling(latex_options = "hold_position", full_width = FALSE)
```

Table 7: Custom Contrasts (Bonferroni Adjusted)

| contrast | estimate | SE | df | t.ratio | p.value |
|---|---|---|---|---|---|
| Poison 1 vs (2+3)/2 | 0.5707 | 0.0799 | 36 | 7.1421 | 0 |
| Treat A vs Treat B | -0.8170 | 0.1066 | 36 | -7.6680 | 0 |